COMPARATIVE LIFE CYCLE ASSESSMENT AND PRODUCT CARBON FOOTPRINT OF CONVENTIONAL, FLY ASH-BASED AND GGBS BASED CONCRETE

A cradle-to-gate study using openLCA

Master's Project Report

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Abstract

This project investigates the greenhouse-gas impacts of three concrete mixtures: conventional Portland-cement concrete, fly ash-blended concrete and ground granulated blast-furnace slag (GGBS)-blended concrete. A comparative life cycle assessment (LCA) was modelled in openLCA 2.5 using ecoinvent 3.9.1 with the Cut-off system model. The functional unit is 1 m³ of concrete, and the selected impact category is climate change, expressed as Global Warming Potential over 100 years (GWP100) using the IPCC 2021 method.

The foreground recipes are based on Tait and Cheung (2016), supplementary Appendix B. The calculated results are 350.67008 kg CO_2-eq/m³ for conventional concrete, 233.79386 kg CO_2-eq/m³ for fly ash concrete and 150.45055 kg CO_2-eq/m³ for GGBS concrete. The fly ash result is preliminary: the 133 kg fly ash input has no suitable elementary-flow provider in the model, so its environmental burden is excluded. It is not a complete or directly comparable result.

A German ÖKOBAUDAT hard coal fly ash dataset (2023, EN 450-1) was imported and examined. It provides pre-calculated indicators under EF 3.1 but no elementary-flow inventory exchanges for recalculation using IPCC 2021. Its GWP-total of 0.013493071 kg CO_2-eq/kg is therefore reported separately; for 133 kg this equals approximately 1.79458 kg CO_2-eq. This estimate is not added to the IPCC 2021 results. Contribution analysis identifies Portland cement as the dominant modelled contributor in Mix 1 and in the incomplete Mix 2 model. In Mix 3, cement and GGBS are the two largest contributors.

Keywords: life cycle assessment; concrete; carbon footprint; openLCA; ecoinvent; fly ash; GGBS; GWP100.

Abbreviations

Term	Meaning
A1–A3	Product stage modules: raw material supply, transport and manufacturing, subject to dataset coverage
EF	Environmental Footprint method
EPD	Environmental Product Declaration
GGBS	Ground Granulated Blast-furnace Slag
GWP100	Global Warming Potential over 100 years
IPCC	Intergovernmental Panel on Climate Change
LCA	Life Cycle Assessment
LCI	Life Cycle Inventory

PCF	Product Carbon Footprint
RoW	Rest of World
SCM	Supplementary Cementitious Material

1. Introduction

1.1 Background

Concrete is a widely used construction material. Its environmental profile depends on the materials and processes involved in its production. Portland cement is an important constituent of conventional concrete and is commonly a major source of greenhouse-gas emissions in concrete supply chains. Replacing a portion of Portland cement with supplementary cementitious materials (SCMs), such as fly ash or GGBS, is one possible strategy for changing the environmental profile of concrete.

The effect of such substitution must be evaluated using a consistent life cycle framework. The result depends not only on the recipe but also on the environmental burdens assigned to each material, the representativeness of background datasets, transport and processing assumptions, system boundaries and impact assessment methods.

1.2 Problem statement

This project compares three concrete mixtures with different binder compositions. A key modelling challenge was finding suitable background inventory datasets for all ingredients. A suitable elementary-flow inventory provider for the fly ash input could not be identified in the selected ecoinvent database. This data gap makes the Mix 2 calculation incomplete and limits the conclusions that can be drawn from it.

1.3 Aim and objectives

The aim is to model and compare the climate-change impacts of conventional, fly ash-blended and GGBS-blended concrete per cubic meter, identify the main modelled contributors and evaluate the effect of background-data limitations.

- Reproduce the reference recipes from Tait and Cheung (2016), Table 1 and Supplementary Appendix A.
- Create three foreground concrete production processes in openLCA 2.5.
- Connect foreground inputs to available ecoinvent 3.9.1 Cut-off datasets where suitable providers exist.
- Calculate GWP100 using the IPCC 2021 method.
- Identify material contribution hotspots.

- Investigate the German ÖKOBAUDAT fly ash dataset and assess its compatibility with the selected impact method.
- Document assumptions, data gaps and limitations.

1.4 Research questions

- How do the calculated GWP100 results compare across the three modelled mixtures?
- Which modelled inputs contribute most to each result?
- How does the missing fly ash provider affect the interpretation of Mix 2?
- How can the ÖKOBAUDAT result be reported without mixing EF 3.1 and IPCC 2021 results?

1.5 Scope

The functional unit is 1 m³ of concrete. The intended boundary is cradle-to-gate (A1–A3), with actual coverage dependent on the selected background datasets. The use phase and end-of-life are outside the current study. The comparison assumes equivalent specified technical performance as its intended basis, following the reference-study framing; no independent mechanical test evidence has been provided to verify equivalent performance.

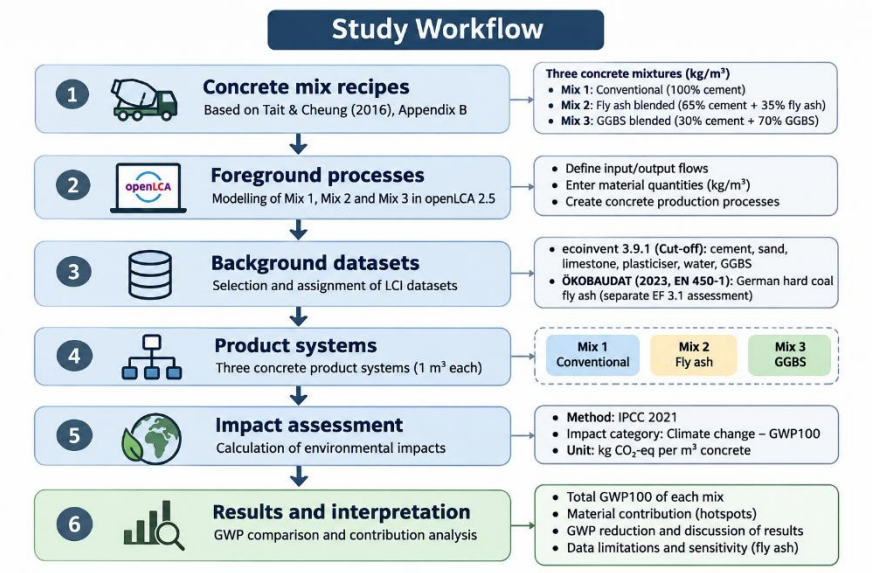


Figure 1. Study workflow for the comparative life cycle assessment of conventional, fly ash-based and GGBS-based concrete using openLCA 2.5. Source: Author's own illustration using AI.

2. Literature Review

2.1 Concrete constituents

The modelled mixtures contain Portland cement, limestone aggregate, sand, water and plasticiser. The blended mixtures additionally contain fly ash or GGBS as supplementary cementitious materials.

2.2 Fly ash and GGBS

Fly ash is a fine material associated with coal combustion and can be used as a supplementary cementitious material when it meets relevant technical requirements. GGBS is a ground material derived from granulated blast-furnace slag and can also replace part of Portland cement in blended binders. Their environmental burdens depend on the inventory, processing, transport and allocation assumptions used. Neither material should automatically be treated as having zero impact.

2.3 LCA framework

LCA is commonly organised into four phases: goal and scope definition, life cycle inventory analysis, life cycle impact assessment and interpretation. In this project, the recipes and linked background datasets form the inventory; IPCC 2021 GWP100 is the selected impact assessment; and the contribution trees and data-gap analysis support interpretation. Cite ISO 14040 and ISO 14044.

2.4 Climate change indicator

The selected indicator is climate change – global warming potential over 100 years (GWP100), expressed in kg CO_2-equivalent. The three openLCA product systems use IPCC 2021. The ÖKOBAUDAT dataset reports pre-calculated indicators under EF 3.1, so its GWP-total is not methodologically interchangeable with the IPCC 2021 results.

2.5 Reference study

The foreground mix recipes are based on Tait and Cheung (2016), "A comparative cradle-to-gate life cycle assessment of three concrete mix designs". The reference study used SimaPro 8 and the ecoinvent 3.01 database. Its functional unit was 1 m^3 of concrete, and its system boundary was cradle-to-gate. The impact assessment methods included Eco-indicator 99, EPD 2008 and Ecopoints 97. The mix compositions were taken from Table 1 and Supplementary Appendix A. In the present study, these recipes are used as the basis for a separate assessment conducted in openLCA 2.5, using ecoinvent 3.9.1 Cut-off and IPCC 2021 GWP100. Therefore, the database and impact assessment methods differ from those used in the reference study.

3. Goal and Scope Definition

3.1 Goal

The goal is to estimate and compare the climate-change impacts associated with producing 1 m³ of three concrete mixtures and identify their main modelled material contributors. The results are for academic analysis and are not verified product-specific environmental declarations.

Parameter	Tait & Cheung (2016)	Present study
Software	SimaPro 8	openLCA 2.5
Database	ecoinvent 3.01	ecoinvent 3.9.1 Cut-off
Functional unit	1 m³ concrete	1 m³ concrete
Boundary	Cradle-to-gate	Cradle-to-gate, intended A1–A3
Impact methods	Eco-indicator 99, EPD 2008, Ecopoints 97	IPCC 2021 GWP100
Geography	UK context	Germany intended; some RoW and European proxies
Recipe source	Table 1 and Supplementary Appendix A	Adapted from reference recipes

Table 1. Comparison of the reference study and the present LCA study.

3.2 Functional unit

The functional unit is 1 m³ of concrete. Equivalent specified technical performance is the intended basis of comparison. This is an assumption based on the comparative framing of the reference study; no independent mechanical testing is included in the information available for this report.

3.3 Product systems

System	Description	Binder composition
Mix 1	Conventional concrete	380 kg Portland cement
Mix 2	Fly ash-blended concrete	247 kg Portland cement + 133 kg fly ash (65:35)
Mix 3	GGBS-blended concrete	114 kg Portland cement + 266 kg GGBS (30:70)

Table 1. Description and binder composition of the three concrete mixtures considered in the comparative life cycle assessment. Source: Adapted from Tait and Cheung (2016).

3.4 System boundary

The intended boundary is cradle-to-gate, corresponding to product-stage modules A1–A3. The foreground process represents the concrete recipe for 1 m³, while upstream supply is represented by linked background datasets where available. The actual coverage of transport and manufacturing depends on the selected dataset documentation. Concrete plant energy, additional transport or other processes not represented by the selected providers must be stated as exclusions or data gaps.

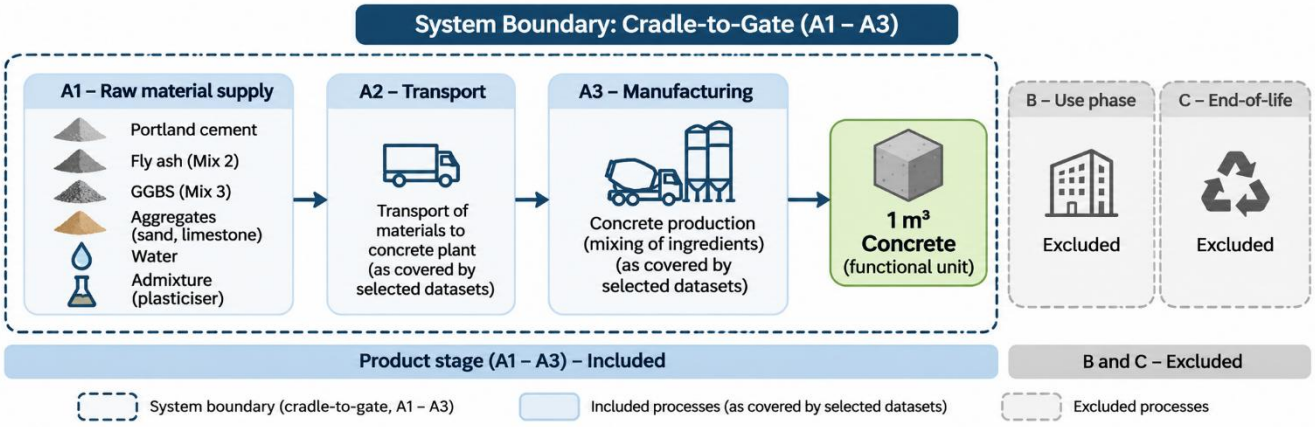


Figure 2. System boundary of the comparative concrete life cycle assessment, illustrating the included product-stage modules A1–A3 and excluded use and end-of-life stages. Source: generated with OpenAI ChatGPT (2026).

3.5 Geography and database

The study is intended to represent a German context, but not all providers are Germany-specific. The selected Portland cement and tap-water providers are Europe without Switzerland; selected sand, limestone and GGBS providers are RoW. These proxies reduce geographical representativeness.

The background database is ecoinvent 3.9.1 using the Cut-off system model. Cite the official database documentation for system-model assumptions and allocation conventions.

3.6 Impact assessment

The calculated indicator is climate change – GWP100 using the IPCC 2021 method selected in openLCA. Results are reported in kg CO_2-equivalent per functional unit.

4. Life Cycle Inventory

4.1 Concrete recipes

The quantities below reproduce the foreground recipes used in the models. All material quantities are kg per m³.

Material (kg/m³)	Mix 1	Mix 2	Mix 3
Portland cement	380	247	114
Fly ash	0	133	0
GGBS	0	0	266
Limestone aggregate	1028	1013	1020
Sand	806	794	800
Plasticiser	2	2	2
Water	190	190	190
Total binder	380	380	380

Table 2. Material quantities (kg/m³) of the three concrete mixtures used in the comparative life cycle assessment. Source: Adapted from Tait and Cheung (2016), with quantities implemented in the author's openLCA models.

4.2 Background datasets and providers

The following provider names and geographies were recorded from the modelling work. Copy the full exact names from openLCA before submission, and verify the plasticiser geography.

Input	Recorded provider / dataset	Geography / note
Portland cement	market for cement, Portland	Europe without Switzerland
GGBS	market for ground granulated blast furnace slag	RoW
Limestone	limestone production, crushed, washed	RoW

Sand	market for sand	RoW
Plasticiser	market for plasticiser based on sulfonated melamine formaldehyde	Verify exact geography
Tap water	market for tap water	Europe without Switzerland
Fly ash	No suitable elementary-flow provider assigned	133 kg remains in foreground; burden omitted

Table 3. Background life cycle inventory datasets and provider geographies assigned to the material inputs in the openLCA models. Source: Author's own modelling based on ecoinvent 3.9.1 Cut-off database.

4.3 Geographical proxies

Where a Germany-specific dataset was not available or selected, regional or RoW datasets were used as proxies. These choices permit calculation but reduce geographical representativeness. Production technologies, energy mixes and supply chains may differ between regions.

4.4 Fly ash dataset investigation

A suitable fly ash elementary-flow inventory provider was not identified in ecoinvent 3.9.1 Cut-off during the project search. Similar-looking entries such as hard coal ash waste flows or fly ash brick production were not treated as equivalent to ready-to-use fly ash for concrete because their product identity and system boundaries differ.

A German ÖKOBAUDAT dataset named 'Hard coal fly ash' was imported. It is identified as Germany (DE), reference year 2023, EN 450-1, dataset version 20.26.070. It represents 1 kg of ready-to-use hard coal fly ash and provides an A1–A3 result. The imported dataset contains pre-calculated environmental indicators under EN 15804+A2 / EF 3.1, but no elementary-flow inventory inputs for recalculation under IPCC 2021. It therefore cannot be used as a direct elementary-flow provider in the current IPCC 2021 product system.

The ÖKOBAUDAT result is retained separately as a method-specific estimate. It is not assigned as the Mix 2 fly ash provider and is not added to the IPCC 2021 result.

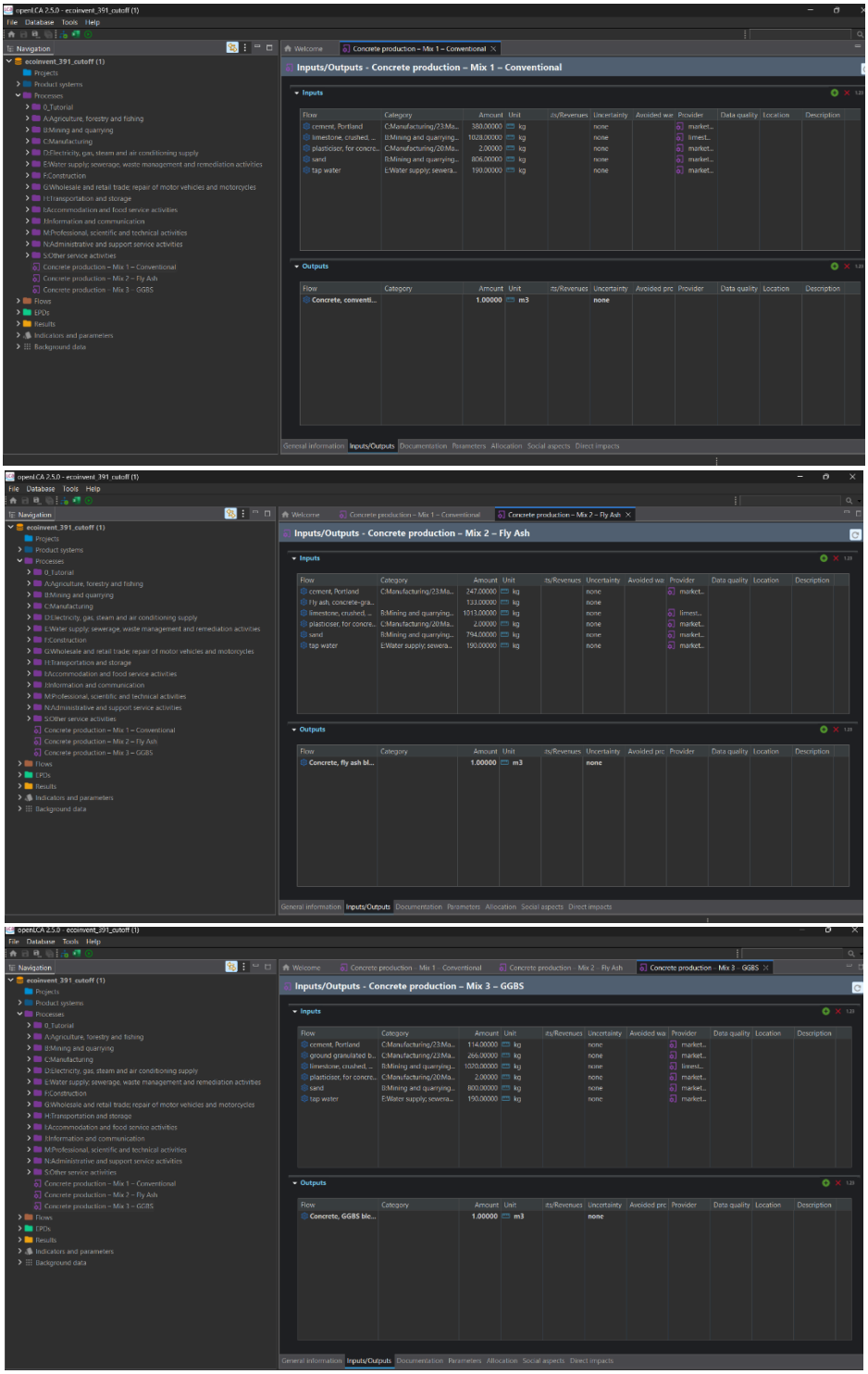


Figure 3. Foreground process inputs and outputs for the three concrete mixtures modelled in openLCA 2.5. Source:openLCA models.

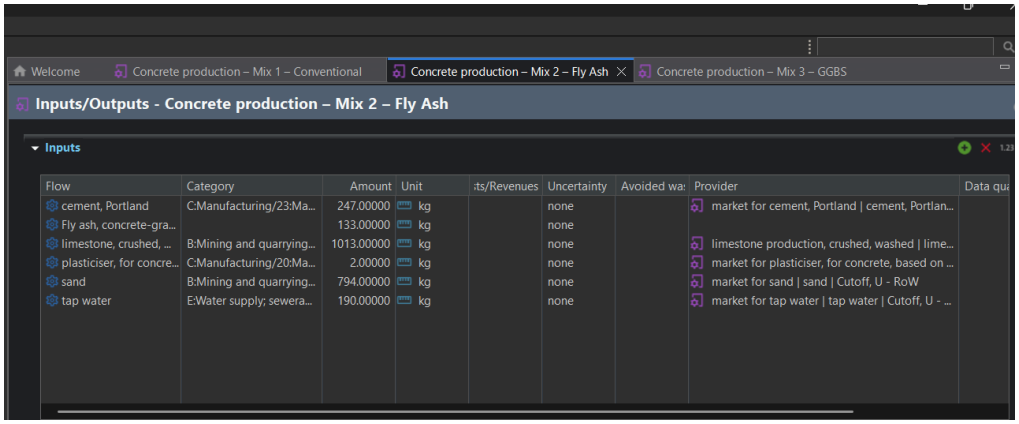


Figure 4. Background dataset assignments for Mix 2, highlighting the unassigned fly ash provider. Source: openLCA model.

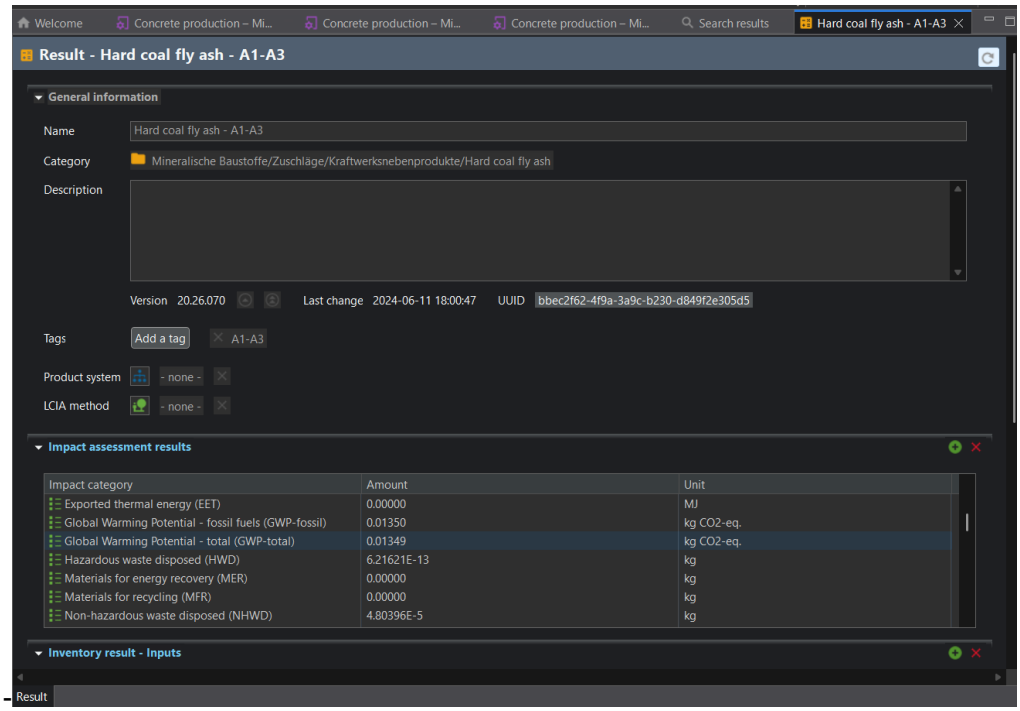


Figure 5. Pre-calculated Global Warming Potential (GWP-total) of German hard coal fly ash for the A1–A3 product stage, showing 0.01349 kg CO_2-eq per kg of fly ash under the EF 3.1 impact assessment method. Source: ÖKOBAUDAT dataset (2023), imported into openLCA 2.5.

5. LCA Modelling in openLCA

5.1 Software and database setup

The foreground processes and product systems were modelled in openLCA 2.5. The background database was ecoinvent 3.9.1 Cut-off. The database contains three concrete production processes and corresponding product systems.

5.2 Foreground processes

Three foreground processes were created: 'Concrete production – Mix 1 – Conventional', 'Concrete production – Mix 2 – Fly Ash' and 'Concrete production – Mix 3 – GGBS'. Each represents the recipe for a reference output of 1 m³ of concrete. Input quantities follow the recipe table in Chapter 4.

5.3 Provider linking

Available background providers were selected for the inputs where suitable datasets could be identified. In Mix 2, the 133 kg fly ash input is present in the foreground process but has no suitable provider. Consequently, the upstream environmental burden of fly ash is not included in the calculated product-system result.

5.4 Product systems and calculation

A product system was generated for each concrete process. Each system was calculated using the IPCC 2021 GWP100 impact assessment method. Contribution trees were reviewed to inspect total results and contributions from linked processes.

5.5 Model verification

The foreground quantities, 1 m³ reference output and provider links were checked. The Mix 2 missing provider was retained as a documented limitation rather than replaced with an unsupported substitute.



Figure 6. Organisation of the three foreground concrete production processes and their corresponding product systems in openLCA 2.5 using the ecoinvent 3.9.1 Cut-off database. Source: Author's own openLCA model.

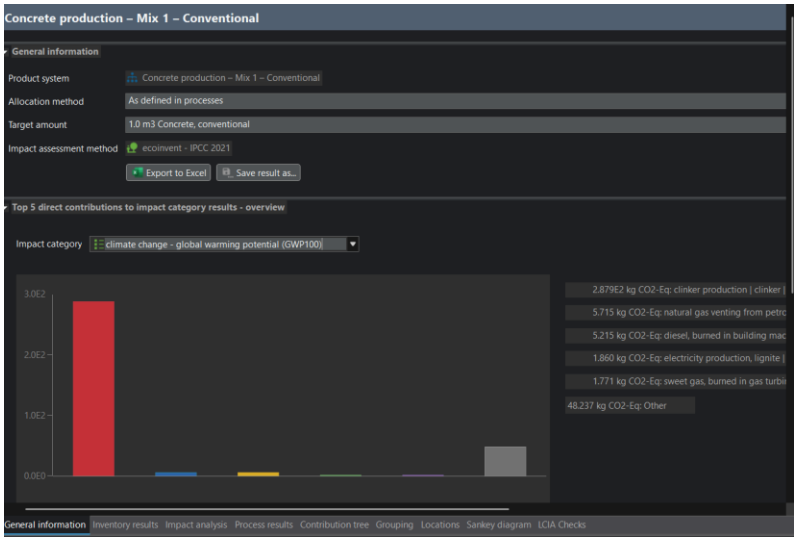


Figure 7. IPCC 2021 impact assessment setup for the conventional concrete product system (Mix 1), showing the functional output of 1 m³ and the Climate Change – Global Warming Potential (GWP100) impact category. Source: openLCA calculation.

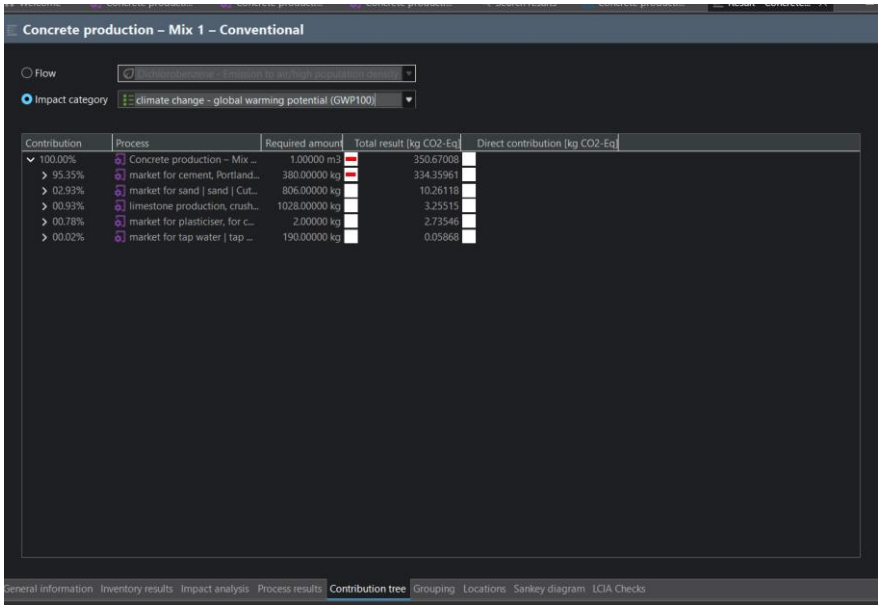


Figure 8. Contribution tree of conventional concrete (Mix 1), showing the contribution of individual material inputs to GWP100. Source: openLCA results.

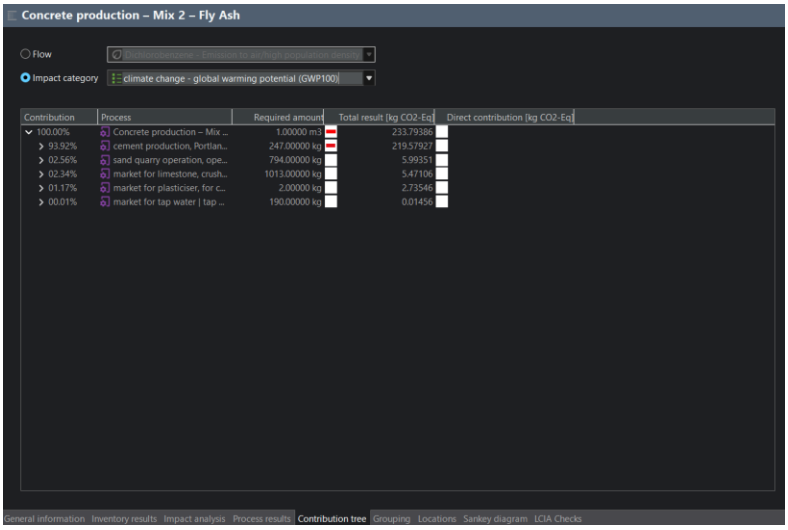


Figure 9. Contribution tree of fly ash-blended concrete (Mix 2), showing the contributions included in the preliminary GWP100 calculation. The environmental burden of 133 kg fly ash is excluded because no suitable elementary-flow provider was assigned. Source:openLCA results.

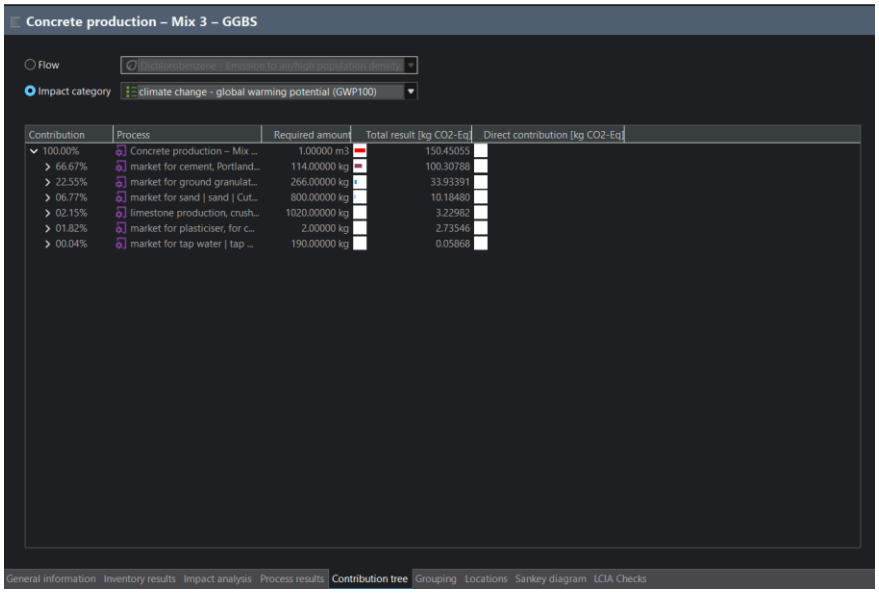


Figure 10. Contribution tree of GGBS-blended concrete (Mix 3), showing the contribution of individual material inputs to GWP100. Source: openLCA results.

6. Results

6.1 Overall GWP100 results

The calculated results are summarised below. Mix 2 is preliminary because the environmental burden of 133 kg of fly ash is excluded from its IPCC 2021 calculation.

Concrete mix	GWP100 (kg CO_2-eq/m^3)	Status
Mix 1 – Conventional	350.67008	Calculated
Mix 2 – Fly ash blended	233.79386	PRELIMINARY – fly ash burden excluded
Mix 3 – GGBS blended	150.45055	Calculated

Table 4. Comparative GWP100 results for the three concrete mixtures, calculated using the IPCC 2021 impact assessment method. Mix 2 is preliminary because the environmental burden of fly ash is excluded. Source: openLCA results.

The numerical Mix 2 result is lower than Mix 1, but it is not a complete comparison because the fly ash burden is missing. The GGBS model result is lower than the conventional model result under the selected datasets and assumptions.

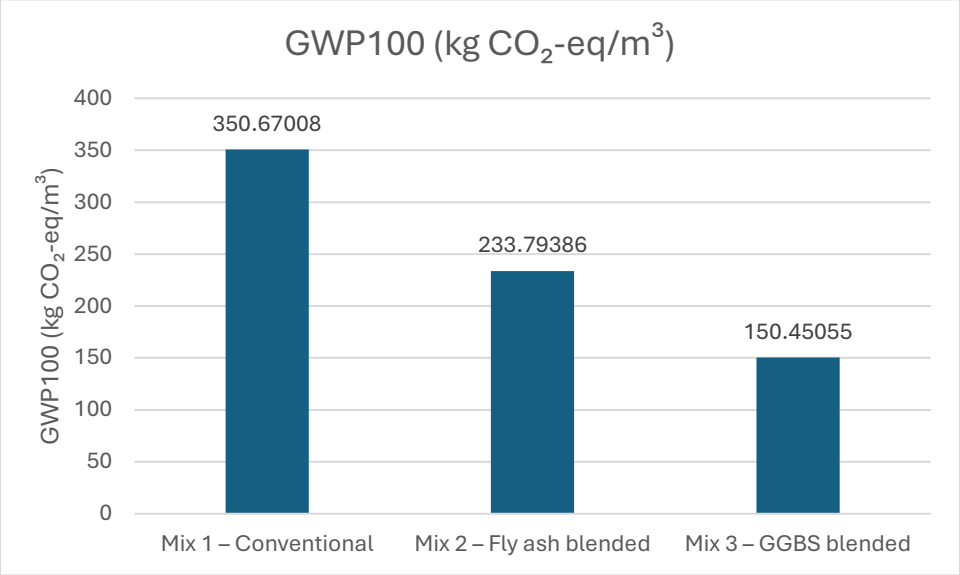


Figure 11. Comparison of GWP100 results for conventional, fly ash-blended and GGBS-blended concrete per 1 m³. The Mix 2 result is preliminary because the environmental burden of fly ash is excluded. Source: LCA results, visualised in Excel.

6.2 Mix 1 contribution analysis

For conventional concrete, Portland cement contributes 334.35961 kg CO_2-eq, or 95.35% of the calculated total. Sand contributes 10.26118 kg CO_2-eq (2.93%), limestone 3.25515 kg CO_2-eq (0.93%), plasticiser 2.73546 kg CO_2-eq (0.78%) and water 0.05868 kg CO_2-eq (0.02%). Portland cement is therefore the dominant modelled hotspot.

Mix 1 input	Contribution (kg CO_2-eq)	Share
Portland cement	334.35961	95.35%
Sand	10.26118	2.93%
Limestone	3.25515	0.93%
Plasticiser	2.73546	0.78%
Water	0.05868	0.02%

Table 5. Material-level contributions to the GWP100 impact of conventional concrete (Mix 1), showing the absolute contribution and percentage share of each input material. Source:openLCA results.

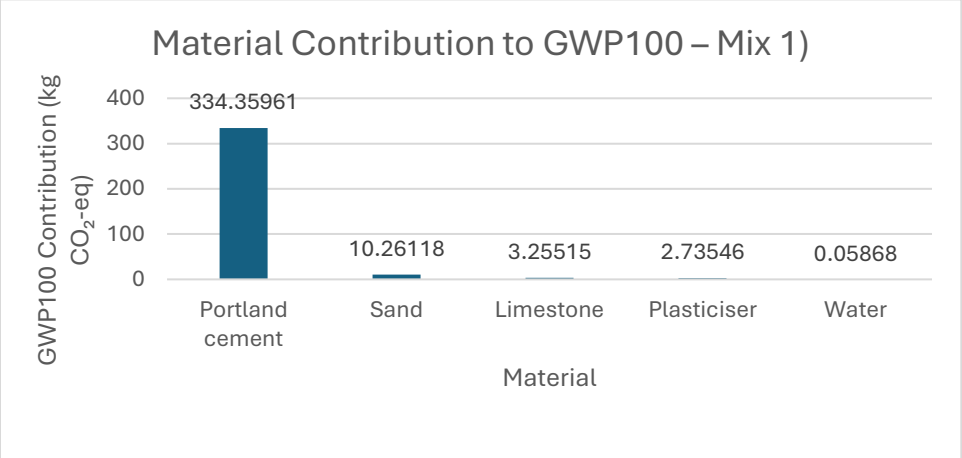


Figure 12. Material hotspot analysis of conventional concrete (Mix 1), illustrating the contribution of individual material inputs to the total GWP100 impact. Portland cement is the dominant contributor, accounting for 95.35% of the total. Source;openLCA results, visualised in Excel.

6.3 Mix 2 contribution analysis – preliminary

The Mix 2 product system reports 233.79386 kg CO_2-eq/m³. Its modelled contributions are Portland cement 219.57927 kg CO_2-eq (93.92%), sand 5.99351 kg CO_2-eq (2.56%), limestone 5.47106 kg CO_2-eq (2.34%), plasticiser 2.73546 kg CO_2-eq (1.17%) and water 0.01456 kg CO_2-eq (approximately 0.01%). The 133 kg fly ash input has no upstream provider and therefore has no modelled contribution.

These percentages describe only the connected part of the Mix 2 model. The absence of a fly ash contribution must not be interpreted as zero environmental impact.

Mix 2 input	Contribution (kg CO_2-eq)	Share of modelled result
Portland cement	219.57927	93.92%
Sand	5.99351	2.56%
Limestone	5.47106	2.34%
Plasticiser	2.73546	1.17%
Water	0.01456	~0.01%
Fly ash	Not modelled	Not available

Table 6. Material-level contributions to the preliminary GWP100 result of fly ash-blended concrete (Mix 2), showing the absolute contributions and percentage shares of the modelled inputs. The fly ash burden is excluded because no suitable elementary-flow provider was assigned. Source;openLCA results.

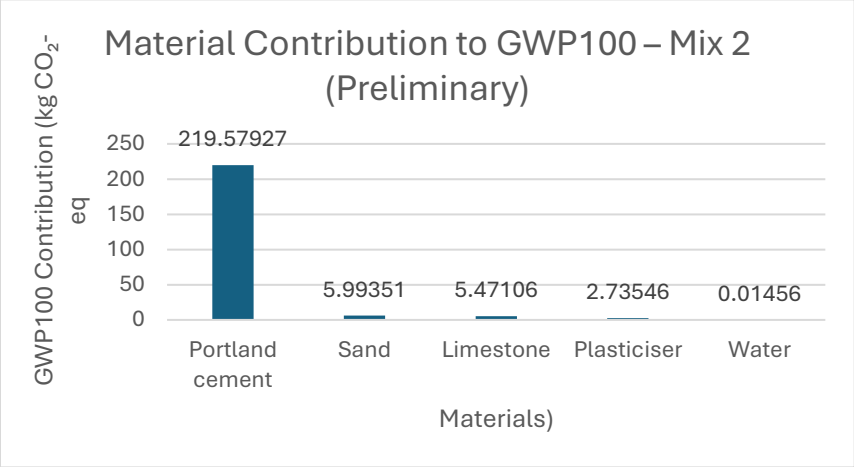


Figure 13. Material contribution analysis of fly ash-blended concrete (Mix 2), showing the contributions included in the preliminary GWP100 assessment. The environmental burden of 133 kg fly ash is excluded because no suitable elementary-flow provider was assigned. Source:openLCA results, visualised in Excel.

6.4 Mix 3 contribution analysis

For the GGBS mixture, Portland cement contributes 100.30788 kg CO_2-eq (66.67%) and GGBS contributes 33.93391 kg CO_2-eq (22.55%). Sand contributes 10.18480 kg CO_2-eq (6.77%), limestone 3.22982 kg CO_2-eq (2.15%), plasticiser 2.73546 kg CO_2-eq (1.82%) and water 0.05868 kg CO_2-eq (0.04%). Cement is the largest modelled contributor, followed by GGBS.

Mix 3 input	Contribution (kg CO_2-eq)	Share
Portland cement	100.30788	66.67%
GGBS	33.93391	22.55%
Sand	10.18480	6.77%
Limestone	3.22982	2.15%
Plasticiser	2.73546	1.82%
Water	0.05868	0.04%

Table 7. Material-level contributions to the GWP100 impact of GGBS-blended concrete (Mix 3), showing the absolute contribution and percentage share of each material input. Source:openLCA results.

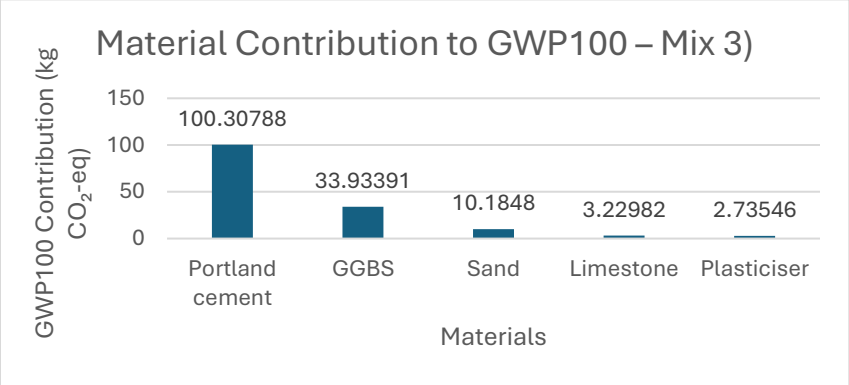


Figure 14. Material hotspot analysis of GGBS-blended concrete (Mix 3), illustrating the contribution of individual material inputs to the total GWP100 impact. Portland cement is the largest contributor (66.67%), followed by GGBS (22.55%). Source:openLCA results, visualised in Excel.

6.5 Reduction comparison

The Mix 3 result is 57.09% lower than Mix 1. The numerical difference between the preliminary Mix 2 result and Mix 1 is 33.34%, but this is not a final reduction estimate because the fly ash burden is absent.

Comparison	Numerical reduction	Status
Mix 2 vs Mix 1	33.34%	Preliminary; fly ash burden excluded
Mix 3 vs Mix 1	57.09%	Based on calculated model results

Table 8. Comparison of GWP100 reductions for fly ash-blended (Mix 2) and GGBS-blended concrete (Mix 3) relative to conventional concrete (Mix 1). The Mix 2 reduction is preliminary because the fly ash environmental burden is excluded. Source:calculations based on openLCA results.

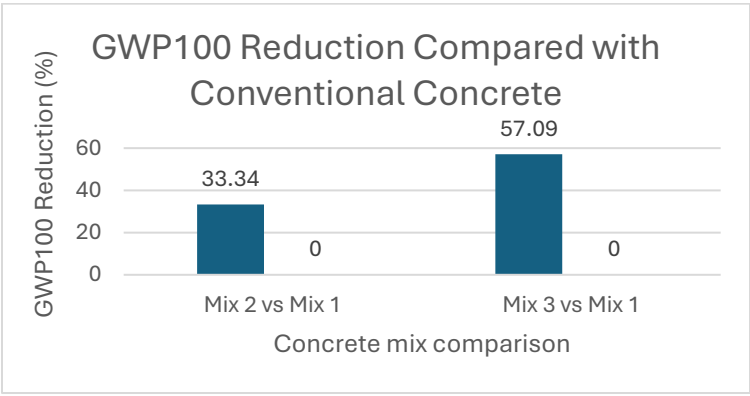


Figure 15. Numerical GWP100 reduction of fly ash-blended (Mix 2) and GGBS-blended concrete (Mix 3) relative to conventional concrete (Mix 1). The Mix 2 reduction is preliminary because the environmental burden of fly ash is excluded. Source:calculations based on openLCA results, visualised in Excel.

6.6 Separate ÖKOBAUDAT EF 3.1 estimate

The ÖKOBAUDAT hard coal fly ash dataset reports GWP-total of 0.013493071 kg CO_2-eq/kg under EF 3.1. For 133 kg of fly ash, the separate estimate is:

133 kg × 0.013493071 kg CO_2-eq/kg = 1.79458 kg CO_2-eq

This estimate is specific to EF 3.1 and the ÖKOBAUDAT dataset. It is reported separately and is not added to the IPCC 2021 Mix 2 result. Combining different impact assessment methods would not produce a consistent IPCC 2021 result.

Parameter	Value	Unit
Fly ash quantity	133	kg
GWP-total per kg fly ash (EF 3.1)	0.013493	kg CO_2-eq/kg
Total GWP for 133 kg fly ash	1.79458	kg CO_2-eq
Formula for the total:		
133 x 0.013493071 =1.79458		

Figure 16. Separate calculation of the GWP-total contribution associated with 133 kg of fly ash using the ÖKOBAUDAT hard coal fly ash dataset under the EF 3.1 impact assessment method. The result is reported separately and is not incorporated into the IPCC 2021 Mix 2 calculation. Source: calculation based on ÖKOBAUDAT dataset results.

7. Discussion

7.1 Conventional concrete

The conventional model is dominated by Portland cement, which accounts for 95.35% of calculated GWP100. Within the selected model and datasets, cement is the principal hotspot. This indicates that binder composition is an important factor to investigate in lower-carbon concrete design.

7.2 GGBS concrete

The GGBS mixture has a calculated GWP100 of 150.45055 kg CO_2-eq/m³ compared with 350.67008 kg CO_2-eq/m³ for conventional concrete, a modelled reduction of 57.09%. Cement remains the largest contributor in Mix 3, followed by GGBS. Interpretation must consider that the selected GGBS provider is a RoW dataset.

7.3 Fly ash concrete

The preliminary Mix 2 result is incomplete because the 133 kg fly ash input has no elementary-flow provider. Its calculated total includes only connected parts of the model. The apparent 33.34% reduction cannot be treated as a final comparative claim. The separate EF 3.1 ÖKOBAUDAT estimate provides contextual information but cannot be added directly to the IPCC 2021 result.

7.4 Comparison with Tait and Cheung

The recipes are based on Tait and Cheung (2016), but this project should not be expected to reproduce the paper's results exactly. Database version, geography, providers, allocation and system boundary can influence results. Insert exact reference-study values and assumptions only after checking the full paper.

7.5 Implications for industry

The project demonstrates an LCA workflow relevant to product sustainability: define a functional unit, collect a bill of materials, connect supply-chain datasets, calculate an impact indicator and identify hotspots. For companies, this can support environmental screening and material-selection decisions. Such decisions require complete and representative inventories; the Mix 2 limitation demonstrates why missing data must be disclosed.

8. Limitations and Data Quality

8.1 Missing fly ash inventory

The principal limitation is the absence of a suitable elementary-flow provider for fly ash in Mix 2. The foreground process contains 133 kg, but its environmental burden is not included. Mix 2 is therefore incomplete and not directly comparable as a final result with Mix 1 and Mix 3.

8.2 EF 3.1 versus IPCC 2021

The imported ÖKOBAUDAT dataset contains pre-calculated EF 3.1 indicators rather than elementary-flow inventory exchanges for recalculation under IPCC 2021. Its GWP-total is retained separately.

The German ÖKOBAUDAT hard coal fly ash dataset (2023, EN 450-1) was imported successfully. However, it provides pre-calculated environmental indicators under EF 3.1 and does not contain elementary-flow inventory exchanges. Therefore, it could not be directly

integrated into the IPCC 2021 impact assessment. Its GWP-total was retained separately for sensitivity analysis.

8.3 Geographical representativeness

RoW and Europe-without-Switzerland datasets are proxies rather than fully Germany-specific datasets. Regional differences in technology, energy and supply chains may influence results.

8.4 Technical performance

Equivalent specified performance is the intended comparison basis, but no independent mechanical test data have been supplied to verify equivalent strength, durability or other performance properties.

8.5 Boundary completeness and uncertainty

The intended boundary is A1–A3, but actual coverage depends on the linked datasets. Additional plant energy, transport or other processes may be absent or embedded in providers. No quantitative uncertainty analysis is available in the current project evidence; uncertainty is therefore discussed qualitatively.

9. Conclusions and Recommendations

9.1 Conclusions

Three concrete foreground models were created in openLCA using recipes based on Tait and Cheung (2016). Their calculated GWP100 values are 350.67008 kg CO_2-eq/m³ for Mix 1, 233.79386 kg CO_2-eq/m³ for Mix 2 and 150.45055 kg CO_2-eq/m³ for Mix 3. Mix 2 is preliminary because the 133 kg fly ash input has no suitable elementary-flow provider and its burden is excluded.

Portland cement accounts for 95.35% of the Mix 1 result. In Mix 3, Portland cement contributes 66.67% and GGBS contributes 22.55%. The Mix 3 result is 57.09% lower than Mix 1 under the selected modelling assumptions. The numerical 33.34% reduction for Mix 2 is preliminary and must not be presented as a final comparison.

The ÖKOBAUDAT hard coal fly ash dataset was imported and examined. Its EF 3.1 pre-calculated result was retained separately because it cannot be directly integrated into the IPCC 2021 inventory calculation. This highlights the importance of inventory completeness and impact-method consistency.

9.2 Recommendations

- Obtain a suitable, documented elementary-flow inventory for the specific fly ash used before reporting a final Mix 2 footprint.

- Use one consistent impact assessment method for all inputs in a final comparison.
- Prefer Germany-specific or regionally representative datasets where available and document proxies.
- Verify the system-boundary and transport coverage of each provider.
- Obtain technical performance evidence if claiming equivalent concrete performance.
- Perform sensitivity or uncertainty analysis when reliable alternative datasets are available.

9.3 Future work

Future work could develop a complete inventory-based fly ash model, test regionalised supply scenarios, include verified transport and concrete-plant data, and assess additional environmental impact categories. Any extension should maintain consistency in functional unit, boundary, allocation and impact assessment method.

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Use of Artificial Intelligence

ChatGPT (OpenAI) was used as an AI-assisted tool to generate the workflow illustration included in this report and to assist with language refinement and presentation. The author reviewed and verified the modelling decisions, openLCA calculations, data interpretation and conclusions. All final academic content and results remain the responsibility of the author

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